Muhammad Faraz Karim

Ph.D. Candidate in Cybersecurity | Georgia Institute of Technology

Atlanta, GA | faraz.karim1999@gmail.com | LinkedIn | GitHub

Research Profile

My research focuses on securing cyber-physical and real-time computing systems, with complementary work in hardware-aware AI acceleration. I develop fault-tolerant system architectures, study isolation and scheduling for cloud and embedded platforms, and explore efficient machine-learning execution for resource-constrained environments.

Education

Georgia Institute of Technology — Atlanta, GA

Ph.D. in Cybersecurity — Jan. 2023 – expected May 2028

Advanced to candidacy: Nov. 2025 | Advisor: Prof. J. Sukarno Mertoguno

Research areas: Systems security; cyber-physical systems; distributed and real-time computing; hardware-aware AI/ML.

Selected coursework: Advanced Operating Systems, Advanced Computer Networks, Advanced Compilers, Cyber-Physical Systems Security, OS Security, Cryptography, Algorithms, Network Security.

Georgia Institute of Technology — Atlanta, GA

M.S. in Computer Science, Computing Systems — May 2026

Lahore University of Management Sciences (LUMS) — Lahore, Pakistan

B.S. in Computer Science | GPA: 3.8/4.0 — Aug. 2018 – Aug. 2022

Selected coursework: Machine Learning, AI, Data Science, Statistics, Probability, Data Mining, Network Security, Advanced Programming, Algorithms.

Research Experience

Georgia Institute of Technology — CSAFA Laboratory — Atlanta, GA

Graduate Research Assistant | Systems and Security — Jan. 2023 – present

SubprocessBFT++: Real-Time Cloud Security — 2026 – present

- Developing a framework that executes diversified subprocess variants sequentially, comparing crash behavior and outputs to detect memory-corruption attacks before results are externally committed.
- Investigating Xen RTDS scheduling and Unikraft-based execution to adapt replication to available deadline budget and workload pressure; exploring deferred presentation and low-copy isolation.

JBODSP: AI Acceleration Using Idle DSP Clusters — Nov. 2025 – present

Lead Researcher | BIRD Energy Consortium, Phase 2 (U.S. Department of Energy)

- Designed an architecture for reusing idle DSP clusters as configurable AI accelerators, and built a Python architectural prototype for 1D convolution tiling and execution on Intel Arria 10 FPGA configurations.
- Modeled on-chip memory budgets, tiling halos, logical compute-unit distribution, and idealized compute throughput to assess design feasibility.
- Proposed cross-compute-unit dispatch, intra-unit parallelism, and double-buffering as opportunities to improve utilization and potentially hide data-transfer stalls.

SubprocessBFT++: Single-Controller Robotic and Industrial Security — 2023 – 2025

Lead Researcher | BIRD Energy Consortium, Phase 1 (U.S. Department of Energy)

- Developed a subprocess-level cyber-attack-resilience architecture using diversified variants and supervisory wrappers to extend BFT++ principles to single-controller cyber-physical systems.
- Implemented C/C++ control and fault-injection components; evaluated timing, disruption, and recovery behavior through drone simulations and microcontroller-oriented experiments, including multi-scan-cycle effects.

LLM Compression and Hardware-Aware Sparse Inference 2026 – present

- Investigating magnitude-based suppression of attention QKV parameters to reduce LLM computation and memory demand for resource-constrained inference.
- Studying compression-induced error propagation and specialized zero-skipping hardware to avoid computation on suppressed weights.

LLM Security and Vibe-Coding Interfaces

- Investigated prompt-injection and related vulnerabilities in commercial LLM web interfaces; developed *EviLLM*, a browser-based framework for security experiments.
- Built Chrome extension components for HTTP request inspection/modification and user-interface instrumentation using JavaScript/TypeScript, extension APIs, service workers, and IndexedDB.

Secure Autonomy Transfer and Multi-Expert Reinforcement Learning

- Developed pre-trained critic–actor methods for transferring autonomy in robotic systems; implemented Gazebo environments, PyTorch training workflows, and evaluation scripts, reporting 25% faster convergence against the tested baseline.
- Investigated multi-expert methods for transferring pretrained actor knowledge, with experiments reporting over 70% improvement in convergence rate relative to tested untrained actor–critic approaches.

Lahore University of Management Sciences — Internet Security and Privacy Lab Lahore, Pakistan

Undergraduate Research Assistant Jan. 2021 – Dec. 2022

- Developed LLVM-based program-debloating workflows and explored BERT-driven specialization-policy selection, including program-analysis datasets and modified compiler passes.
- Built a benchmarking and correctness-testing pipeline for CHISEL, OCCAM, RAZOR, and PIECEWISE; evaluated code reduction, with some experiments removing more than 50% of program bloat.
- Led an exploratory statistical analysis of generational patterns in suicide incidence using historical data; published an article on *The Startup*.

Publications

Peer-reviewed

M. F. Karim, Y. Deng, L. Niu, B. Ramasubramanian, M. Alexiou, D. Sahabandu, R. Poovendran, and J. S. Mertoguno. "Rapid Autonomy Transfer in Reinforcement Learning with a Single Pre-Trained Critic." *IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*, 2024. [DOI]

D. R. Keppler, **M. F. Karim**, M. S. Mickelson, and J. S. Mertoguno. "Experimentation and Implementation of the BFT++ Cyber-Attack Resilience Mechanism for Cyber-Physical Systems." *ACM Transactions on Cyber-Physical Systems*, 8(3), article 35, 2024. [DOI]

M. Ali, M. Muzammil, **M. F. Karim**, A. Naeem, R. Haroon, M. Haris, H. Nadeem, W. Sabir, F. Shaon, F. Zaffar, V. Yegneswaran, A. Gehani, and S. Rahaman. "SoK: A Tale of Reduction, Security, and Correctness—Evaluating Program Debloating Paradigms and Their Compositions." *European Symposium on Research in Computer Security (ESORICS)*, pp. 229–249, 2023. [DOI]

Manuscripts under review

M. F. Karim and J. S. Mertoguno. "SubprocessBFT++: Providing the Security Guarantees of BFT++ on a Single-Controller." Under review, IEEE International Conference on Robotics and Automation (ICRA 2027).

E. Ryu*, S. Chung*, **M. F. Karim**, A. Raymaker, K. S. Jodha, Y. Chaturvedi, and J. S. Mertoguno. "Beware EviLLM: Enabling Vulnerability Injection via Large Language Models." Manuscript under review, 2026. *Equal contribution.

Professional Experience

Ibrands ai Texas, USA

Cybersecurity Consultant Sept. 2025 – present

- Advised on cybersecurity controls, client data protection, and security considerations for sensitive-data pipelines.

Brains Inc Pakistan Pakistan

Cybersecurity Compliance Consultant Sept. 2025 – present

- Conducted cybersecurity compliance reviews and on-site assessments.

Educative Seattle, WA (remote)

Software Engineer Advocate Jan. 2021 – Jan. 2022

- Authored 100+ technical tutorials and runnable examples in Python, Kotlin, C/C++, Go, and Rust, spanning network security and backend development.
- Designed interactive coding widgets and diagrams; collaborated through Git-based review workflows.

Selected Technical Projects

ProdeBench: Debloating Benchmarking and Comparison June 2021 – July 2022

- Developed a benchmarking platform integrating CHISEL, OCCAM, RAZOR, and PIECEWISE; automated debloating, metric generation, and correctness checks, including OCCAM workflows.
- Conducted with Dr. Fareed Zaffar in collaboration with Dr. Ashish Gehani and Dr. Sazzadur Rehman.

BERT for Optimizing TRIMMER Oct. 2021 – July 2022

- Modified TRIMMER's LLVM passes for sub-callgraph specialization, prepared a code dataset, and pre-trained a BERT model to investigate data-driven specialization policies.
- Conducted with Dr. Fareed Zaffar in collaboration with Dr. Ashish Gehani and Dr. Hashim Sharif.

Khaaba.me: Homemade-Food Marketplace

- Built a web marketplace using MongoDB, Mongoose, Node.js, React, Redux, Bootstrap, and AWS; performed basic penetration testing and implemented application security controls.

AI Vehicle Identification and Video Tracking

- Developed a camera-video vehicle classifier/tracker and compared SVM, neural-network, logistic-regression, and Naive Bayes models with HOG, PCA, SIFT, and color/texture features.

Generational Patterns in Suicide Rates

- Led analysis of WHO data (1985–2016) using exploratory analysis, hypothesis testing, and logistic regression under Dr. Ihsaan Ayyub Qazi.

Teaching and Mentorship

Curriculum Co-Developer, TinyML Security Georgia Tech / University of Washington, 2024 – present. Developing lab-focused teaching materials on attacks and defenses for machine-learning models on embedded hardware.

Guest Lecturer and Teaching Assistant Introduction to Cyber-Physical Systems Security, Georgia Tech; invited lectures in Feb. and Oct. 2023 and TA appointment Aug.–Dec. 2023 (Prof. Sukarno Mertoguno).

Teaching Assistant Network Security, LUMS, Aug.–Dec. 2022.

Graduate Mentor Lahore Women in Computing — GradAppLab, 2023 – present; supporting prospective graduate applicants.

Ph.D. Welcome Day Research Representative Selected to brief incoming Georgia Tech students on cyber-physical systems security research.

Invited Talks and Presentations

- RSAC, BIRD Energy Consortium Workshop, San Francisco, CA, 2026.
- Invited talk on SubprocessBFT++ and cyber-physical systems security, BIRD Energy Consortium Workshop, Georgia Institute of Technology, Atlanta, GA, 2024.
- Invited talk on SubprocessBFT++ and cyber-physical systems security, BIRD Energy Consortium Workshop, Arizona State University, Tempe, AZ, 2023.

Honors and Awards

- **Finalist**, Foley Scholar PhD Research Impact Award, Georgia Tech, 2026.
- **Service and Engagement Award**, School of Cybersecurity and Privacy, Georgia Tech, 2024.
- **Award of High Distinction**, LUMS, 2022.
- **Dean's Honor List**, LUMS, all semesters, 2018–2022.
- **Merit Scholarship**, LUMS, 2018–2020.

Professional Service and Leadership

Peer Review and Conference Service Program Committee Member and Reviewer, IEEE ICTAI (2026); Reviewer, ACNS Cycle 2 (2026); External Reviewer, AINA (2024); Conference Volunteer, IEEE ICTAI (2024).
Georgia Tech AI Safety Initiative Member, Sept. 2024 – present; discussions on AI safety for autonomous systems and critical-infrastructure applications.
Georgia Tech Ph.D. Visit Day Selected presenter on personal and broader Georgia Tech CPS security research.
JZT Pakistan Member, 2016 – present; organized awareness activities and blood-donor coordination related to genetic blood disorders.
INDEX, LUMS Treasurer, Sept. 2021 – June 2022.
CodinGuru Convener, Sept. 2021 – Feb. 2022; organized a large domestic coding competition.
IEEE LUMS Vice President, Sept. 2020 – June 2021; supported an 80-person membership and 15+ events.
LUMS Badminton Team Vice Captain, Sept. 2020 – Sept. 2021; coordinated training and tournaments.

Technical Skills

Languages Python, C, C++, Go, Rust, Kotlin, JavaScript/TypeScript, SQL, Haskell, R, PHP, LaTeX.
Systems and Infrastructure LLVM, Linux, Xen/RTDS, Unikraft, Docker, AWS (EC2, S3), Git, GitHub Actions, Node.js, Express, REST APIs, MongoDB, PostgreSQL.
Security Tools Wireshark, Burp Suite, Kali Linux, Nmap, Netcat, John the Ripper, Metasploit, OWASP ZAP, Chrome Extensions (MV3).
Machine Learning and Data PyTorch, TensorFlow, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn.
Robotics, Web, and Collaboration ROS2, Gazebo, MuJoCo, PX4, ArduPilot, React, Redux, HTML/CSS, Bootstrap, Figma, Mural, Trello, Slack.